



DATA ANALYST-FINAL PROJECT



CUSTOMER PERSONALITY ANALYST

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DATA UNDERSTANDING AND GOALS

Data: marketing_campaign
Analysis: Phyton & Power BI



DATA UNDERSTANDING



#	Column	Non-Null Count		Dtype
---	-----	-----	-----	----
0	ID	2240	non-null	int64
1	Year_Birth	2240	non-null	int64
2	Education	2240	non-null	object
3	Marital_Status	2240	non-null	object
4	Income	2216	non-null	float64
5	Kidhome	2240	non-null	int64
6	Teenhome	2240	non-null	int64
7	Dt_Customer	2240	non-null	object
8	Recency	2240	non-null	int64
9	MntWines	2240	non-null	int64
10	MntFruits	2240	non-null	int64
11	MntMeatProducts	2240	non-null	int64
12	MntFishProducts	2240	non-null	int64
13	MntSweetProducts	2240	non-null	int64
14	MntGoldProds	2240	non-null	int64
15	NumDealsPurchases	2240	non-null	int64
16	NumWebPurchases	2240	non-null	int64
17	NumCatalogPurchases	2240	non-null	int64
18	NumStorePurchases	2240	non-null	int64
19	NumWebVisitsMonth	2240	non-null	int64
20	AcceptedCmp3	2240	non-null	int64
21	AcceptedCmp4	2240	non-null	int64
22	AcceptedCmp5	2240	non-null	int64
23	AcceptedCmp1	2240	non-null	int64
24	AcceptedCmp2	2240	non-null	int64
25	Complain	2240	non-null	int64
26	Z_CostContact	2240	non-null	int64
27	Z_Revenue	2240	non-null	int64
28	Response	2240	non-null	int64
dtypes: float64(1), int64(25), object(3)				
memory usage: 507.6+ KB				

- 2240 rows and 29 columns.
- 25 columns = integer, 3 columns = object, 1 column = float.
- Dt_customer: The customer registration date with the company → needs to be changed to the appropriate data type.

PROBLEMS & OBJECTIVES

PROBLEMS

Many companies struggle to accurately target customers in their marketing campaigns. Some of the main issues that often arise include:

- Ineffective Marketing Campaigns
- Lack of Clear Segmentation
- Limited Understanding of Customer Behavior

OBJECTIVES

Improving the effectiveness of marketing campaigns, personalizing marketing strategies, and predicting customer responses.





DATA CLEANING & MANIPULATION



DATA CLEANING (1)

MISSING VALUE HANDLING

ID	0
Year_Birth	0
Education	0
Marital_Status	0
Income	24
Kidhome	0
Teenhome	0
Dt_Customer	1324
Recency	0
MntWines	0
MntFruits	0
MntMeatProducts	0
MntFishProducts	0
MntSweetProducts	0
MntGoldProds	0

NumDealsPurchases	0
NumWebPurchases	0
NumCatalogPurchases	0
NumStorePurchases	0
NumWebVisitsMonth	0
AcceptedCmp3	0
AcceptedCmp4	0
AcceptedCmp5	0
AcceptedCmp1	0
AcceptedCmp2	0
Complain	0
Z_CostContact	0
Z_Revenue	0
Response	0

TOTAL

There are 24 missing values in the Income variable and 1,324 missing values in Dt_Customer.

% MISSING

The percentage of missing values:
Income = 1.07%
Dt_Customer = 59.10%

HANDLING MISSING VALUE

- Income = No handling needed.
- Dt_Customer = Drop the variable.



DATA CLEANING (2)

CHECK DUPLICATES

```
[ ] len(df_marketing.drop_duplicates())/len(df_marketing)
```

```
1.0
```

Conclusion: There are no duplicates in the data.



DATA MANIPULATION

CREATE A NEW VARIABLE: FREQUENCY

Create a variable 'Frequency' that comes from Total NumDealsPurchases, NumWebPurchases, NumCatalogPurchases, and NumStorePurchases

CREATE A NEW VARIABLE: MONETARY

Create a variable 'Monetary' that comes from Total MntWines, MntFruits, MntMeatProducts, MntFishProducts, MntSweetProducts, MntGoldProds, and Monetary

DROP UNUSED COLUMN

Column: Z_CostContact, Z_Revenue



	NumDealsPurchases	NumWebPurchases	NumCatalogPurchases	NumStorePurchases	\
0	3	8	10	4	
1	2	1	1	2	
2	1	8	2	10	
3	2	2	0	4	
4	5	5	3	6	
Frequency					
0	25				
1	6				
2	21				
3	8				
4	19				

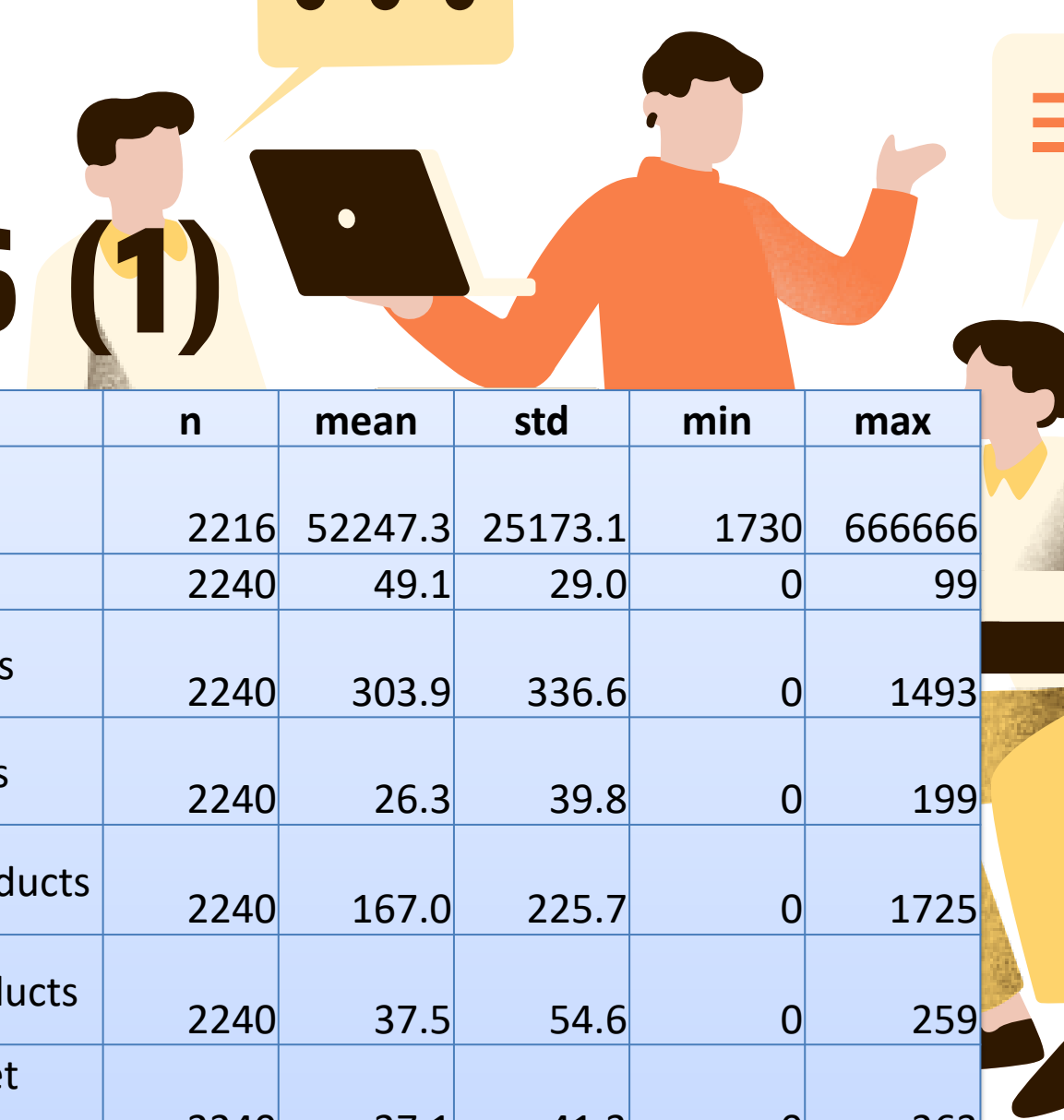
	MntWines	MntFruits	MntMeatProducts	MntFishProducts	MntSweetProducts	\
0	635	88	546	172	88	
1	11	1	6	2	1	
2	426	49	127	111	21	
3	11	4	20	10	3	
4	173	43	118	46	27	
	MntGoldProds	Monetary				
0	88	1617				
1	6	27				
2	42	776				
3	5	53				
4	15	422				



EXPLORATORY DATA ANALYSIS



EXPLORATORY DATA ANALYSIS (1)



OVERVIEW OF NUMERIC VARIABLES

- Income and customer purchase frequency: High variation.
- Wine and meat: Highest average spending.
- Sweets and fruits: Lowest average spending.
- Most frequent purchases: In-store (average: 5.79).
- Least frequent purchases: Catalog (average: 2.66).

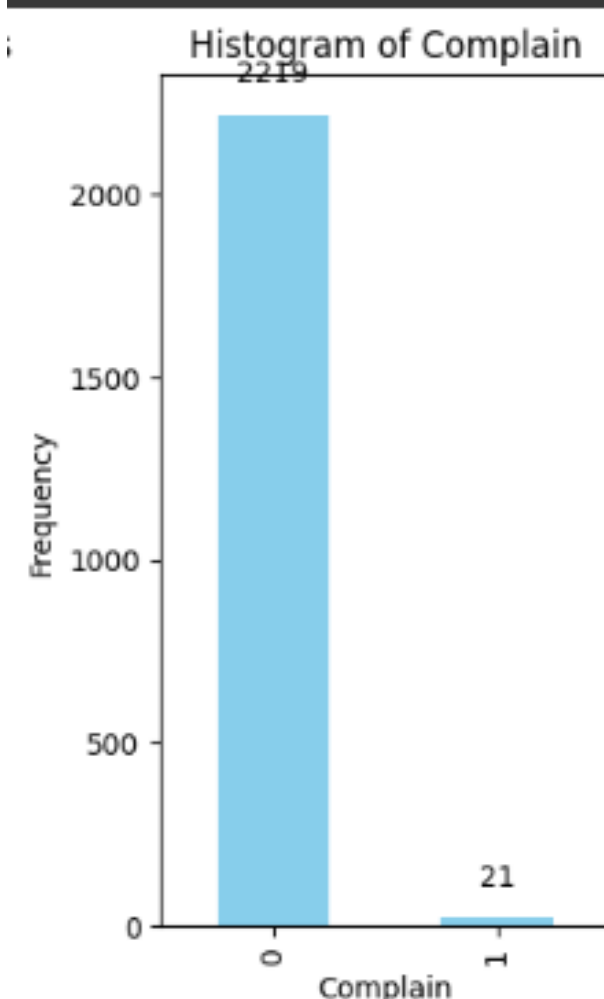
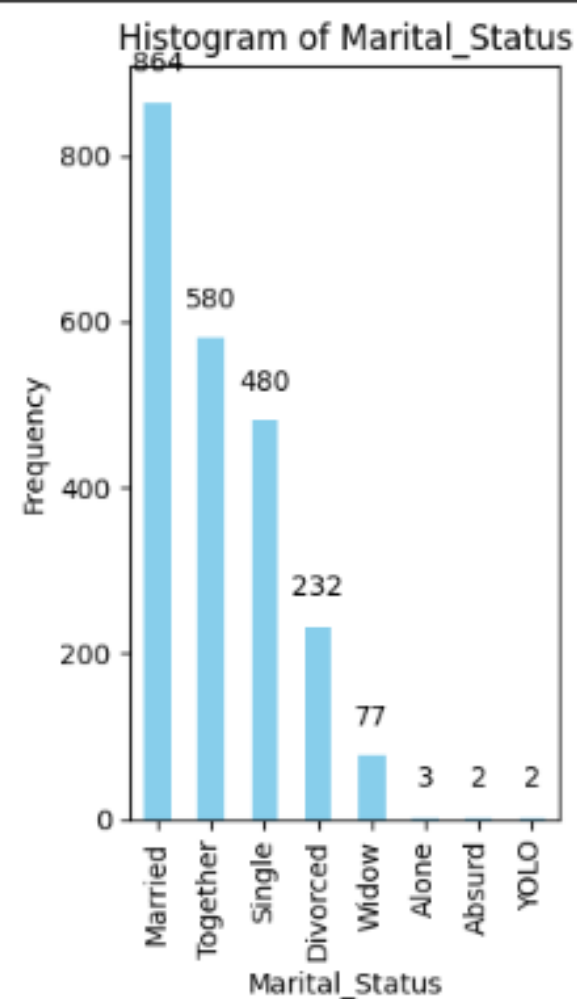
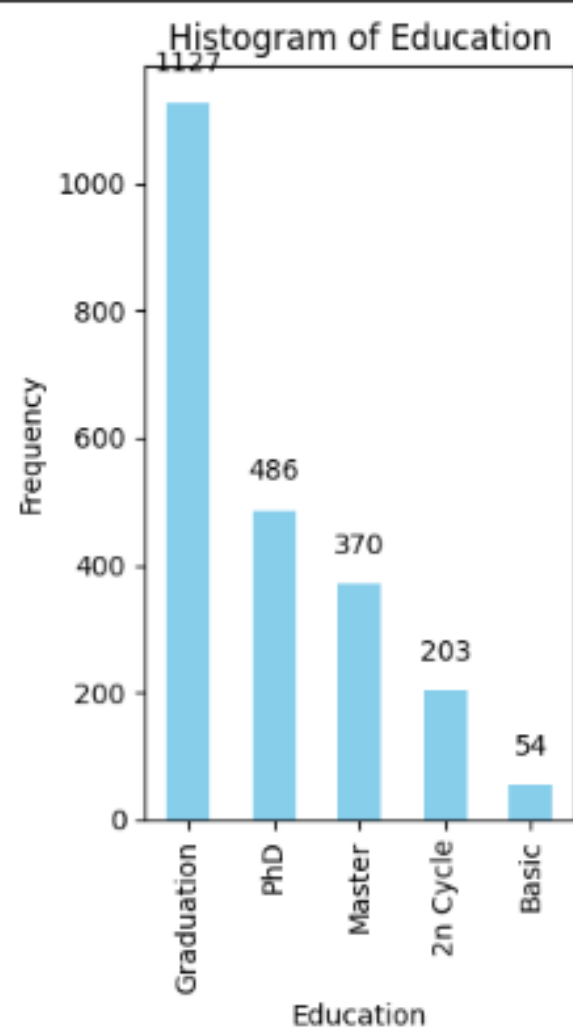
Variable	n	mean	std	min	max
Income	2216	52247.3	25173.1	1730	666666
Recency	2240	49.1	29.0	0	99
Amount spent on Wines	2240	303.9	336.6	0	1493
Amount spent on Fruits	2240	26.3	39.8	0	199
Amount spent on Meat Products	2240	167.0	225.7	0	1725
Amount spent on Fish Products	2240	37.5	54.6	0	259
Amount spent on Sweet Products	2240	27.1	41.3	0	263
Amount spent on Gold Proods	2240	44.0	52.2	0	362
Number of purchases with a discount	2240	2.3	1.9	0	15
Number of Web Purchases	2240	4.1	2.8	0	27
Number of Catalog Purchases	2240	2.7	2.9	0	28
Number of Store Purchases	2240	5.8	3.3	0	13
Number of Web Visits Month	2240	5.3	2.4	0	20
Frequency	2240	14.9	7.7	0	44
Monetary	2240	605.8	602.2	5	2525



EXPLORATORY DATA ANALYSIS (2)

OVERVIEW OF EDUCATION, MARITAL STATUS, AND COMPLAINTS

- Bachelor's and Doctorate: Highest education levels.
- Married and Together: Most common marital statuses.
- No complaints submitted: Highest category (0), indicating high satisfaction.



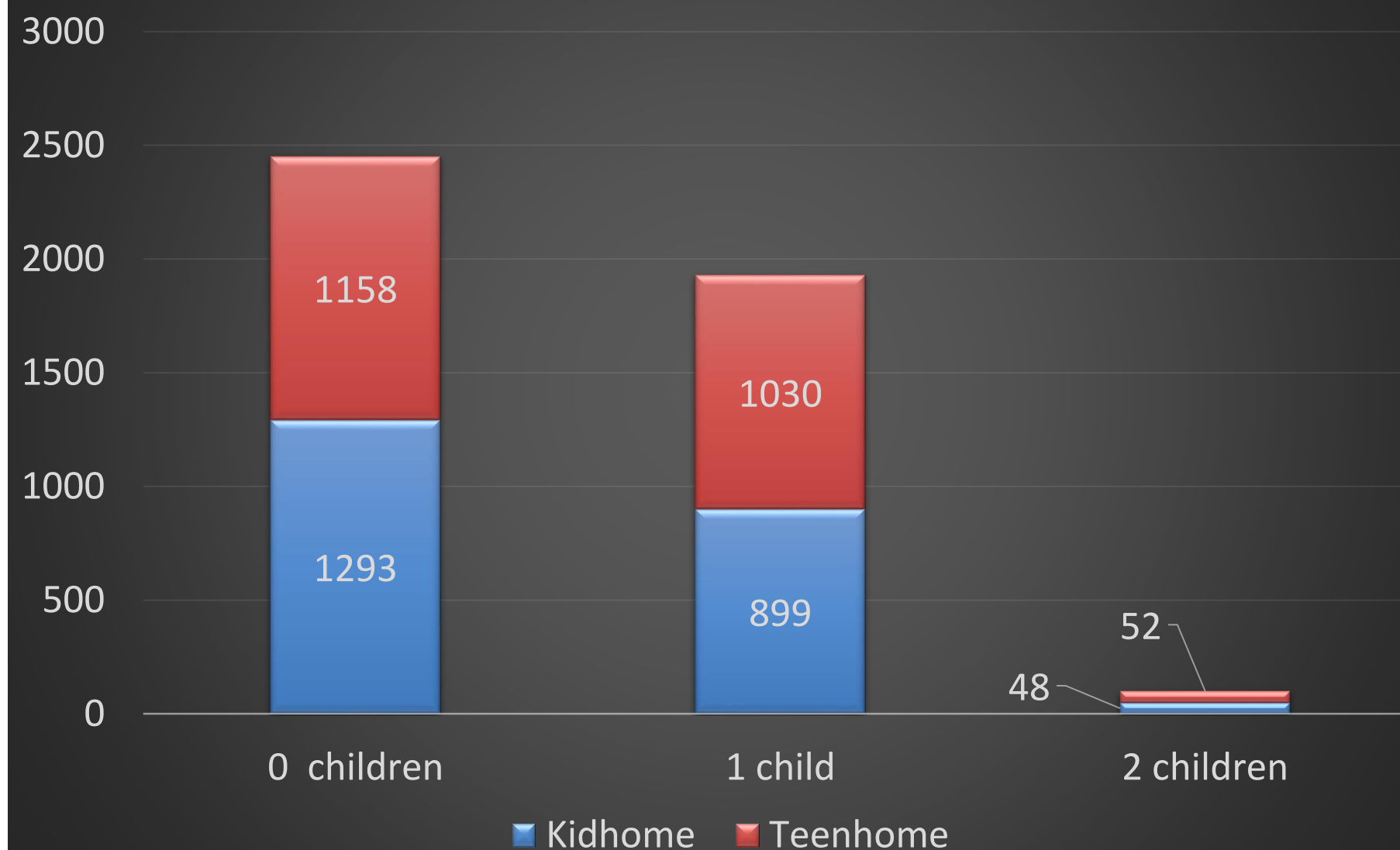


EXPLORATORY DATA ANALYSIS (3)

OVERVIEW OF KIDHOME AND TEENHOME

- The most common category for both Kidhome and Teenhome is having no young children (count = 0). Kidhome: 1,293, Teenhome: 1,158.
- The least common category for both Kidhome and Teenhome is having 2 children. Kidhome: 48, Teenhome: 52.

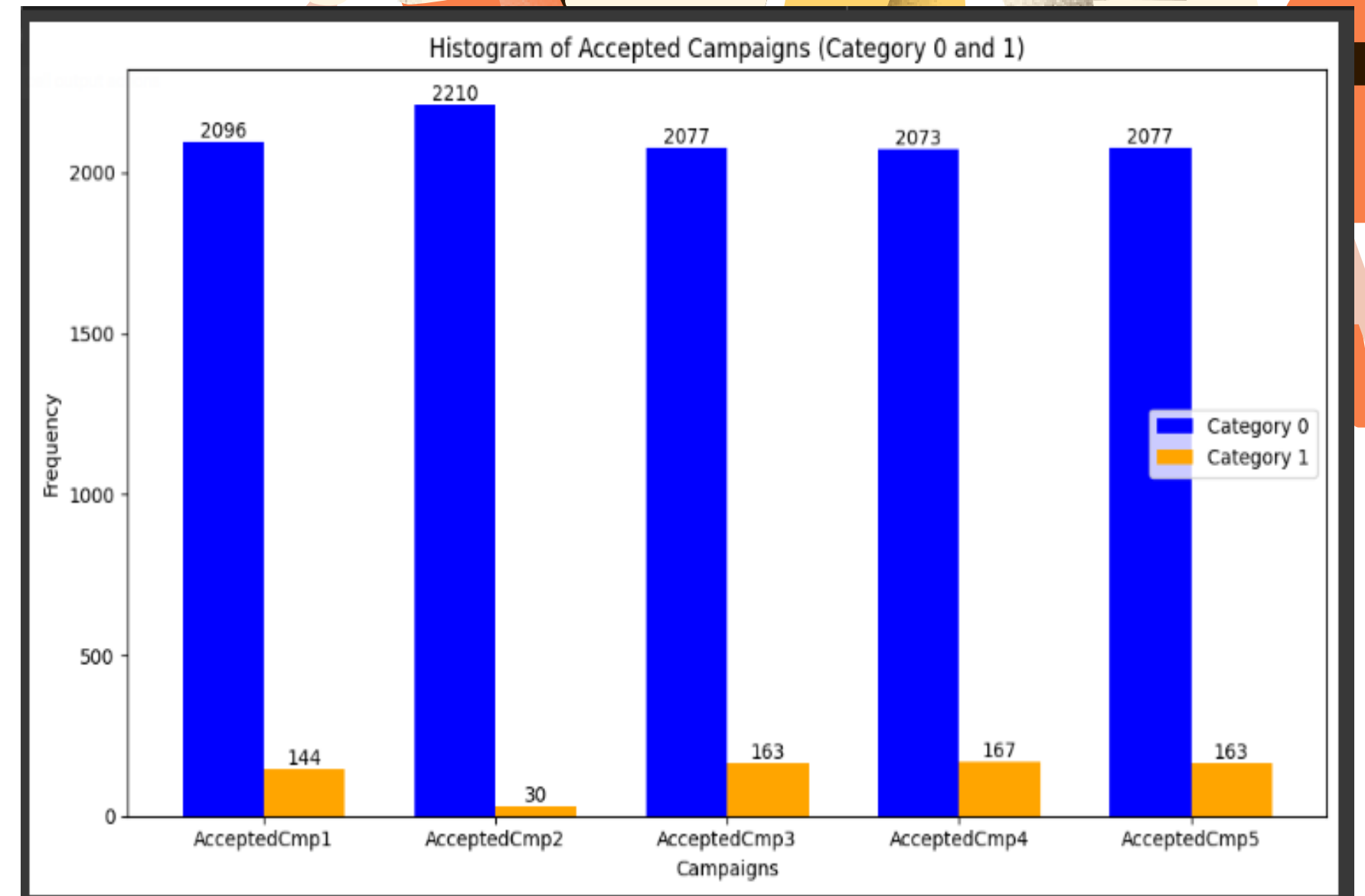
Histogram of Kidhome and Teenhome



EXPLORATORY DATA ANALYSIS (4)

OVERVIEW OF ACCEPTED CAMPAIGNS

- Did not accept any campaign: the largest category, with a frequency of over 2,000 in each campaign.
- Accepted all campaigns: the smallest category, with a frequency below 170.





DASHBOARD

Power BI



Dashboard Marketing Campaign

Education

Select all

2n Cycle

Basic

Graduation

Master

PhD

2240

Count of ID

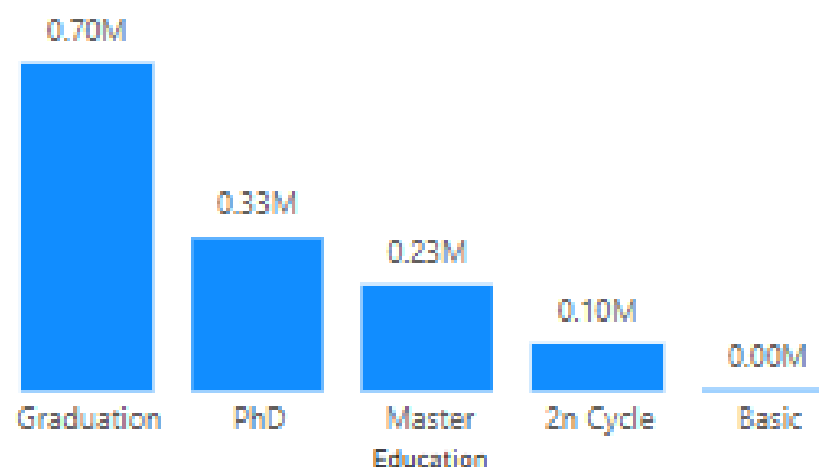
1bn

Total Income

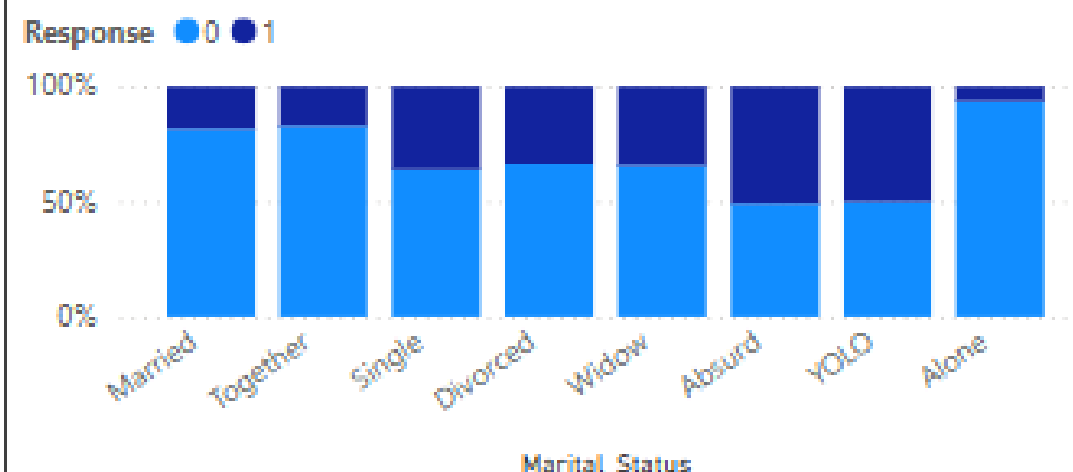
1M

Total Monetary

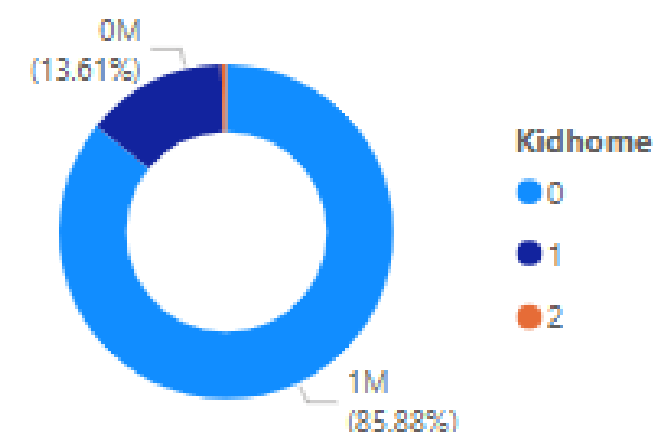
Total Monetary by Education



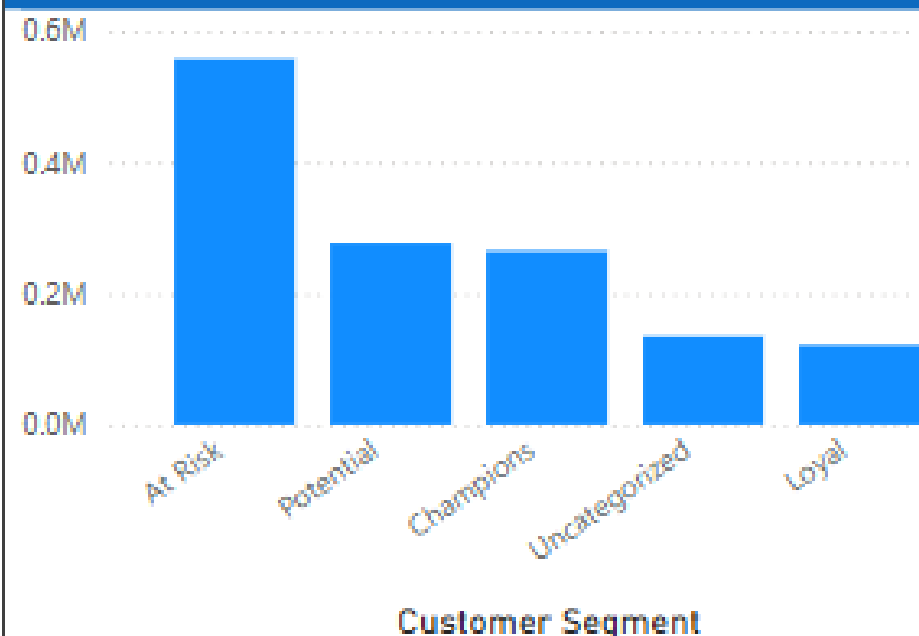
Total Monetary by Marital_Status and Response



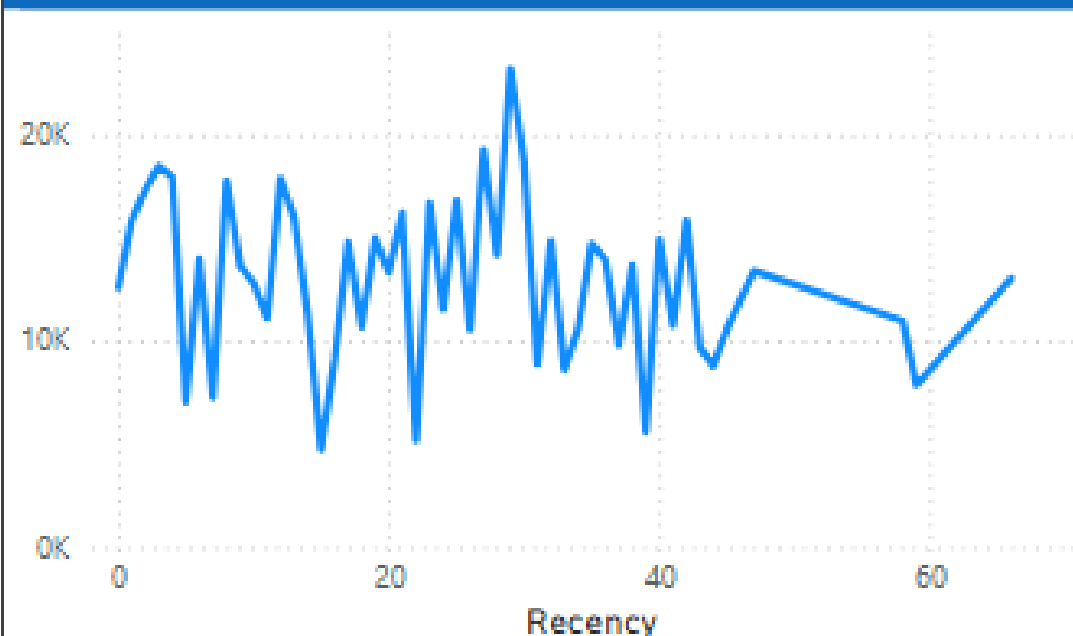
Total Monetary by Kidhome



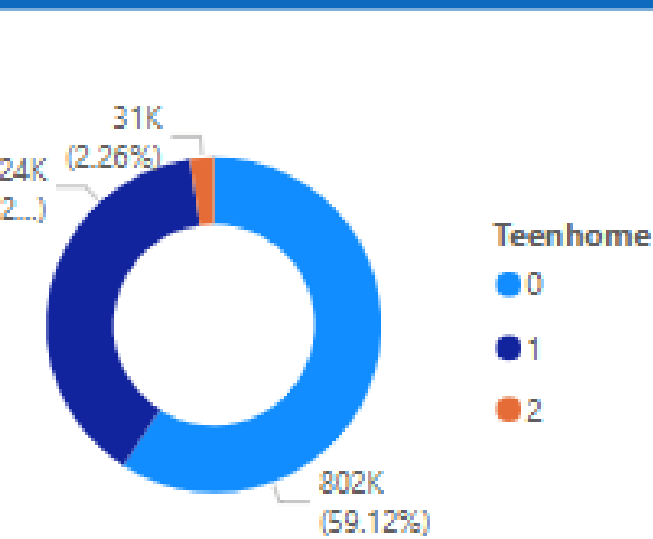
Total Monetary by Customer Segment

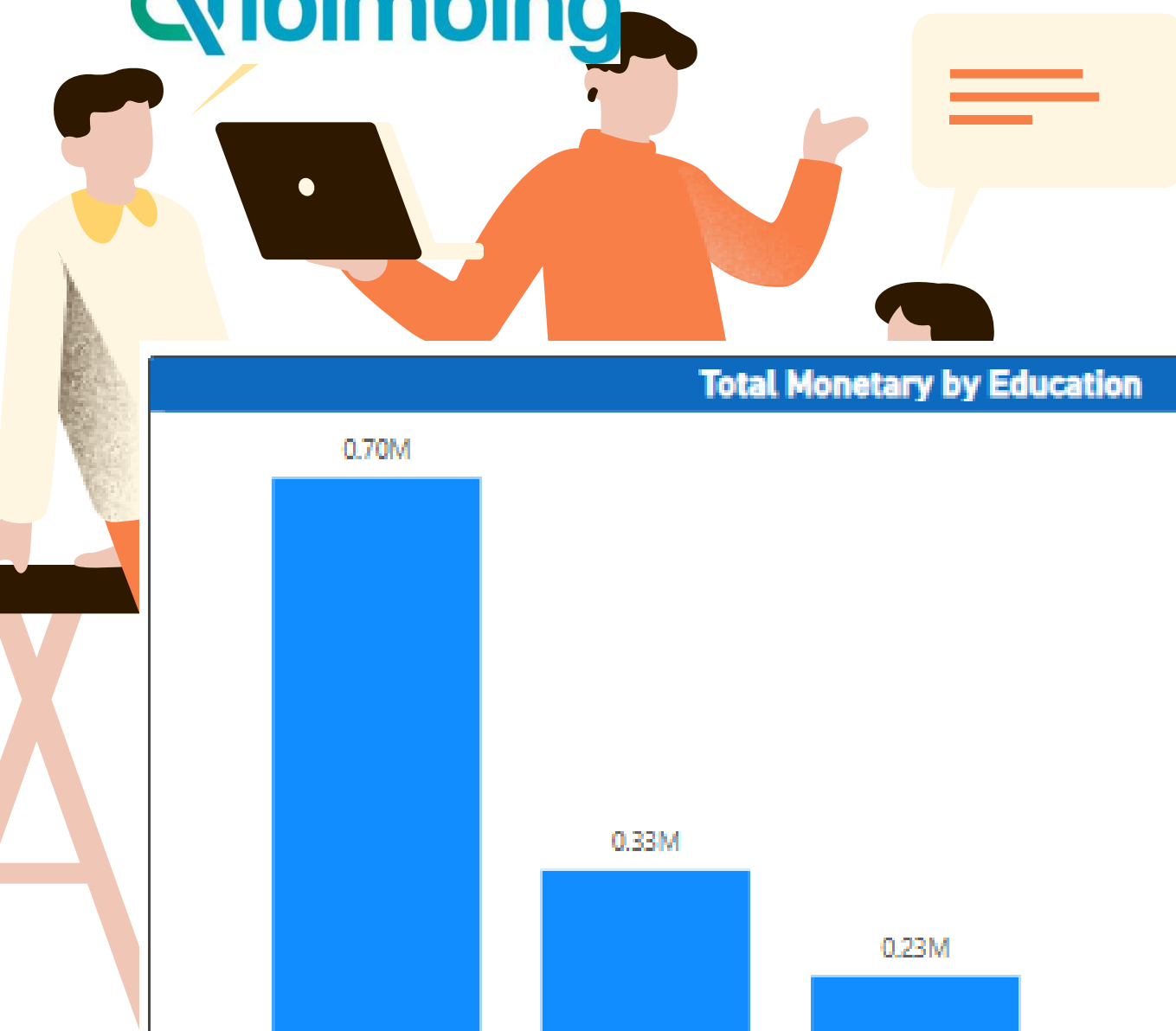


Total Monetary by Recency

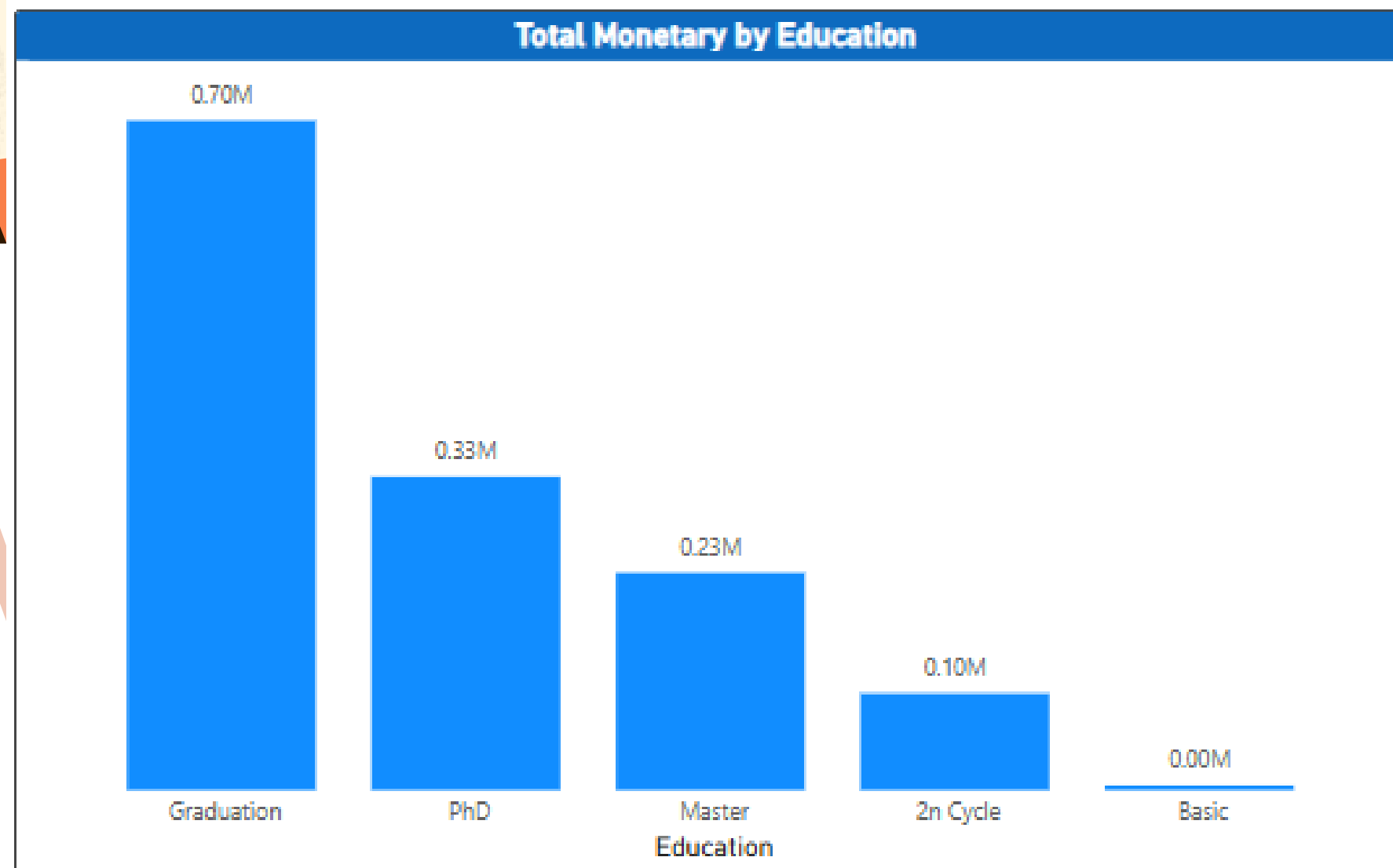


Total Monetary by Teenhome

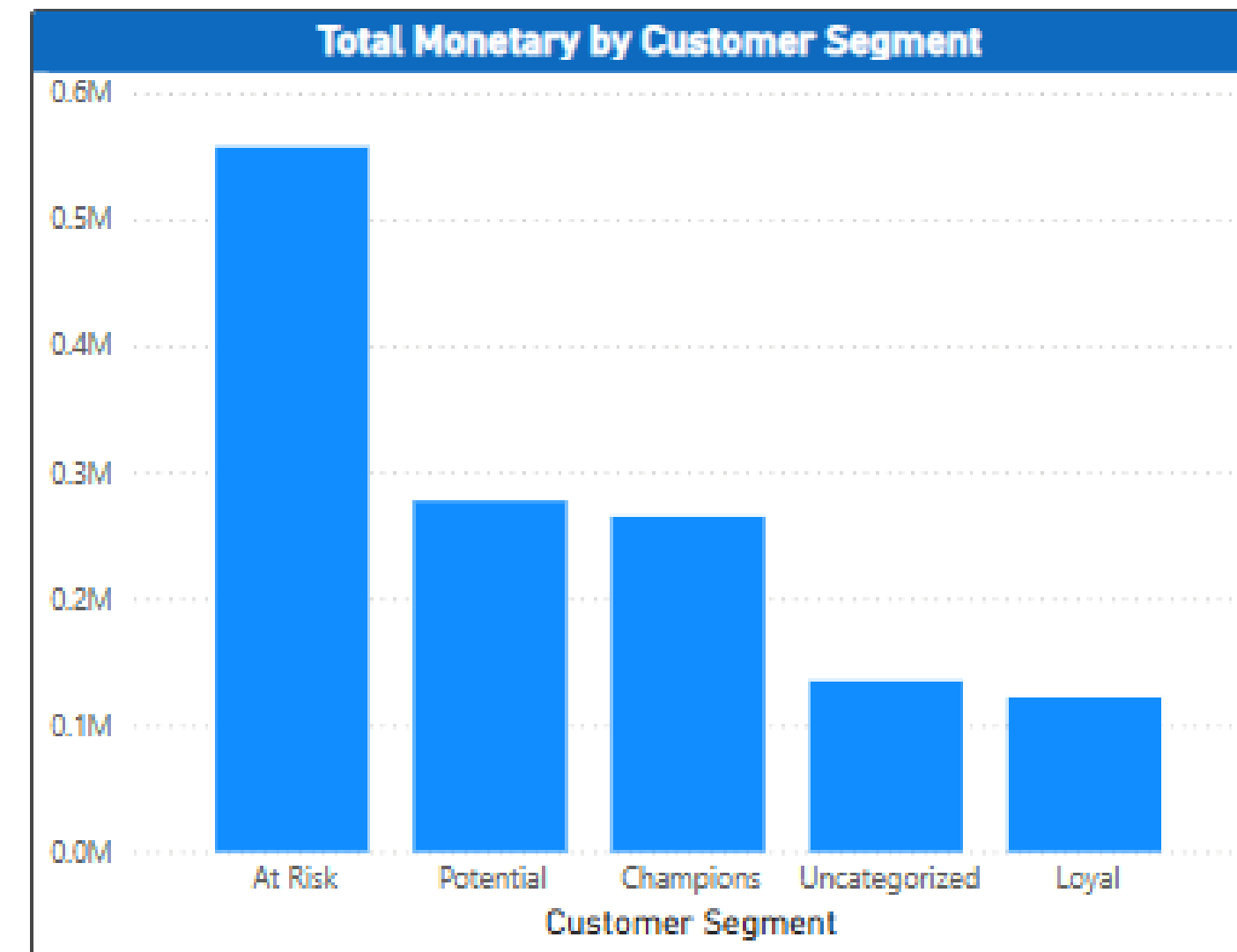




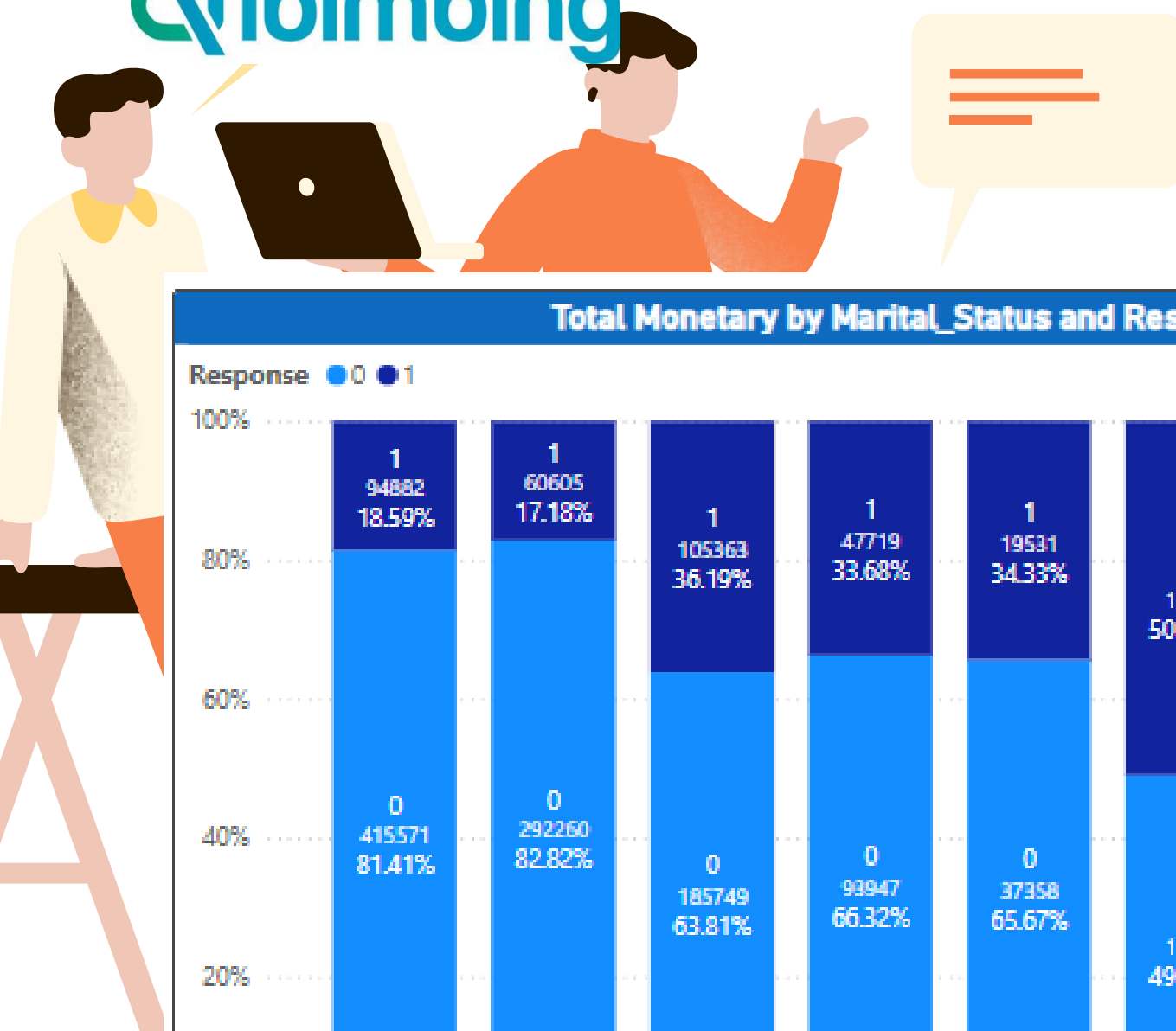
DASHBOARD (1)



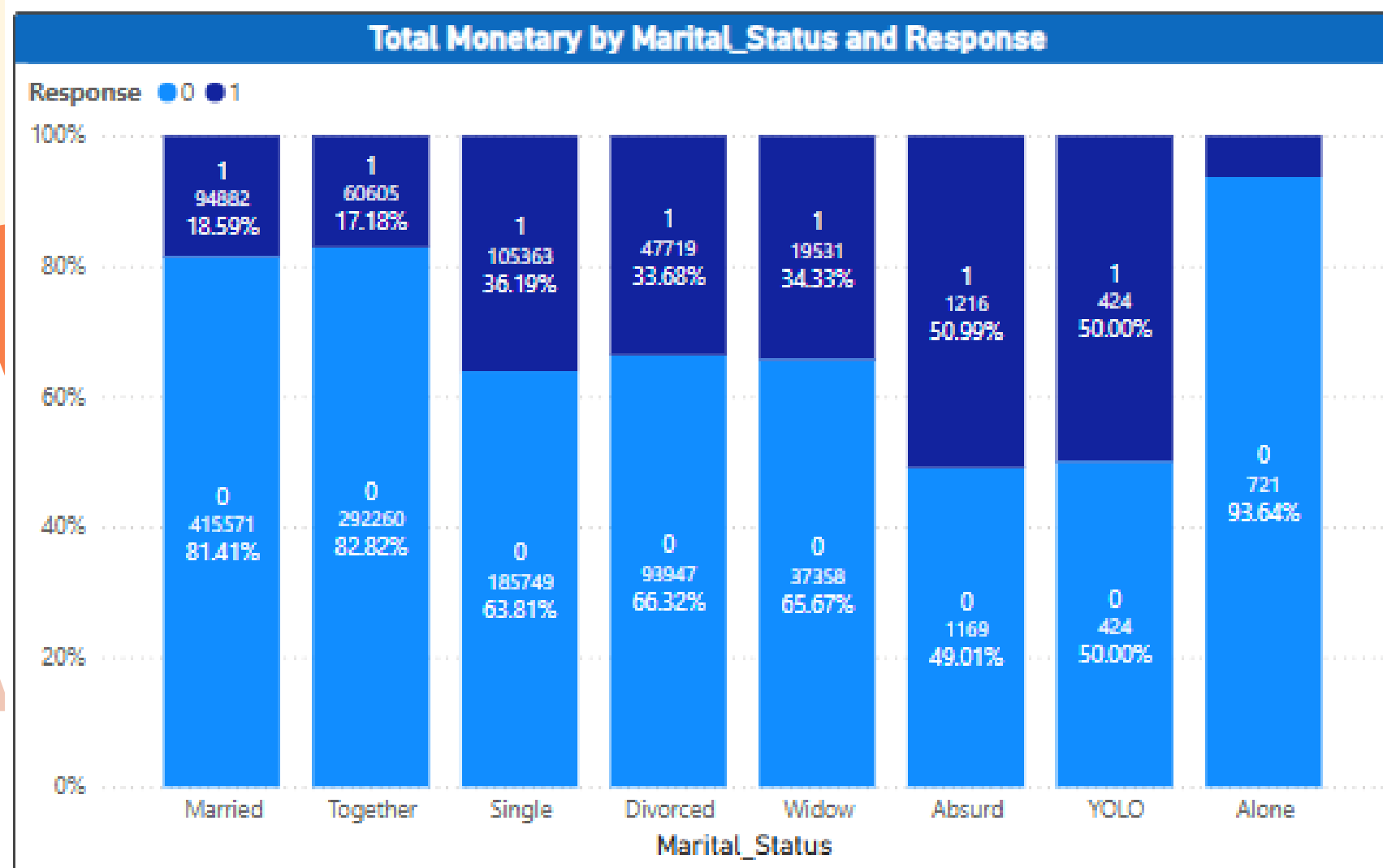
Bachelor's degree: The group with the highest monetary value.
 Basic education: The group with the lowest monetary value.



At Risk: The segment that contributes the most to revenue.
 Loyal: The smallest segment, but very important due to its stability and long-term potential.



DASHBOARD (2)

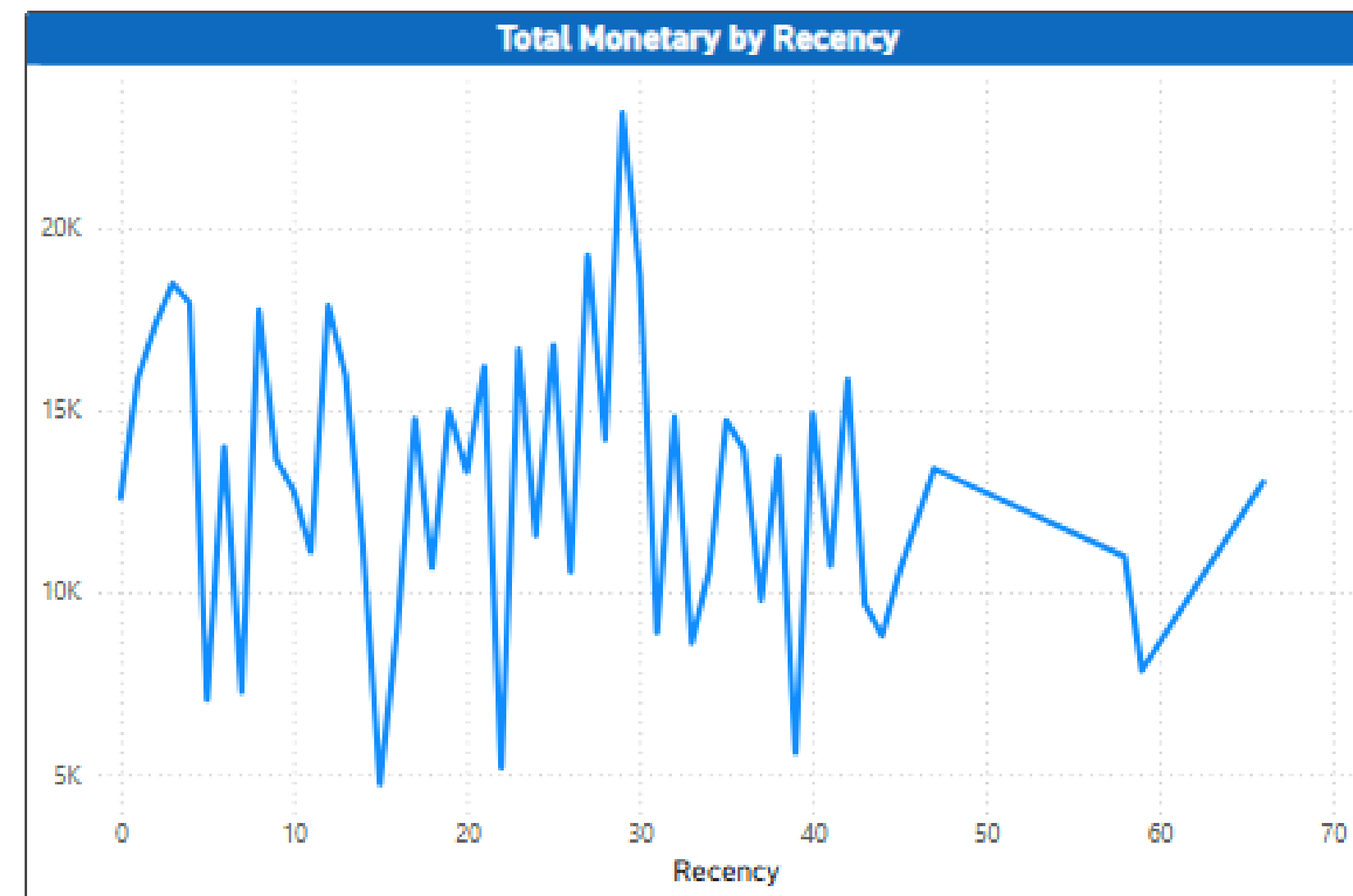


The highest response rate to campaigns: individuals with "Absurd" and "YOLO" status.

The highest non-response rate: "Alone," "Together," and "Married."

The highest total monetary value: married individuals who did not respond, totaling 415,571.

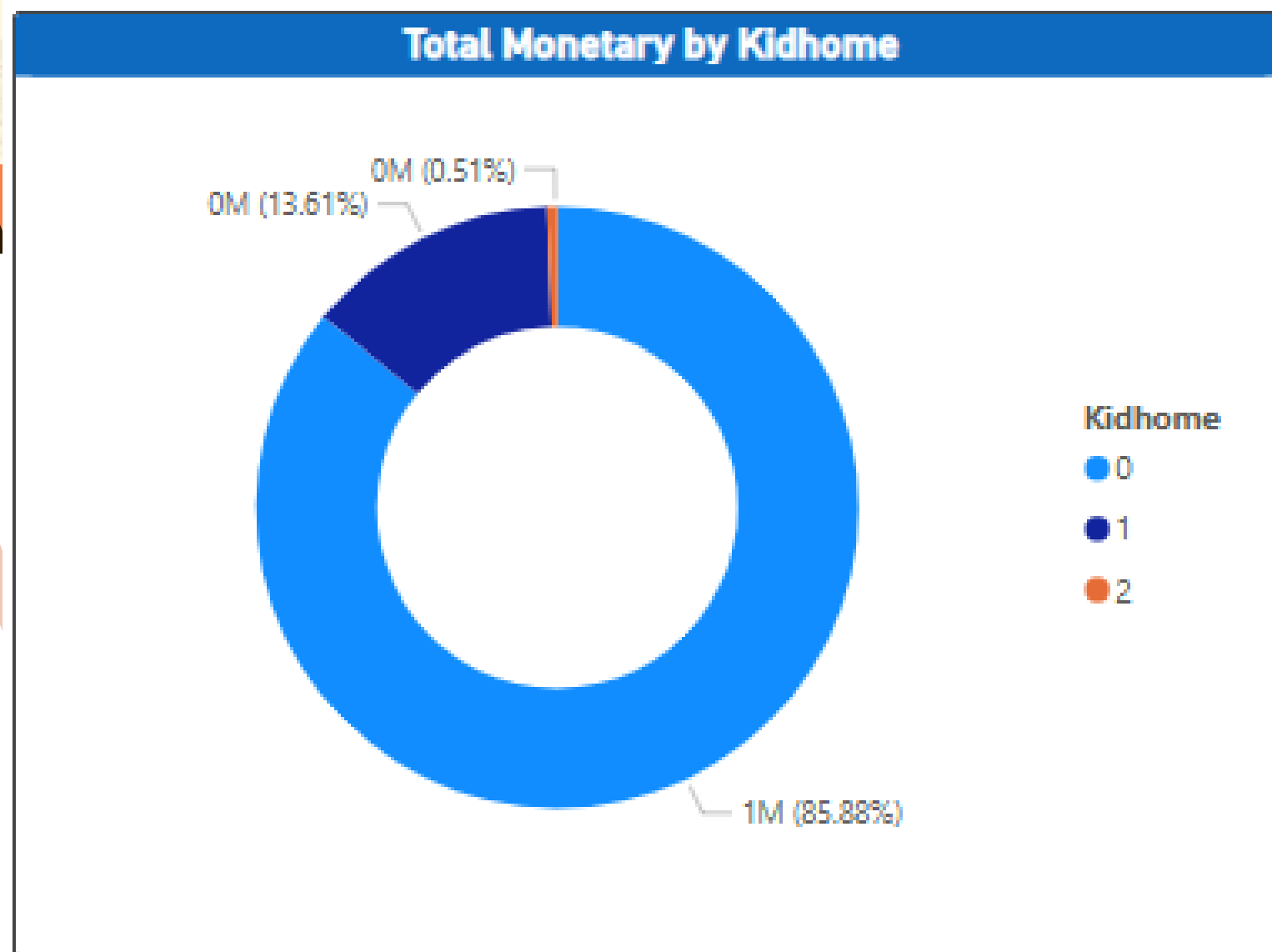
The lowest total monetary value: YOLO, both in response and non-response, totaling 424.



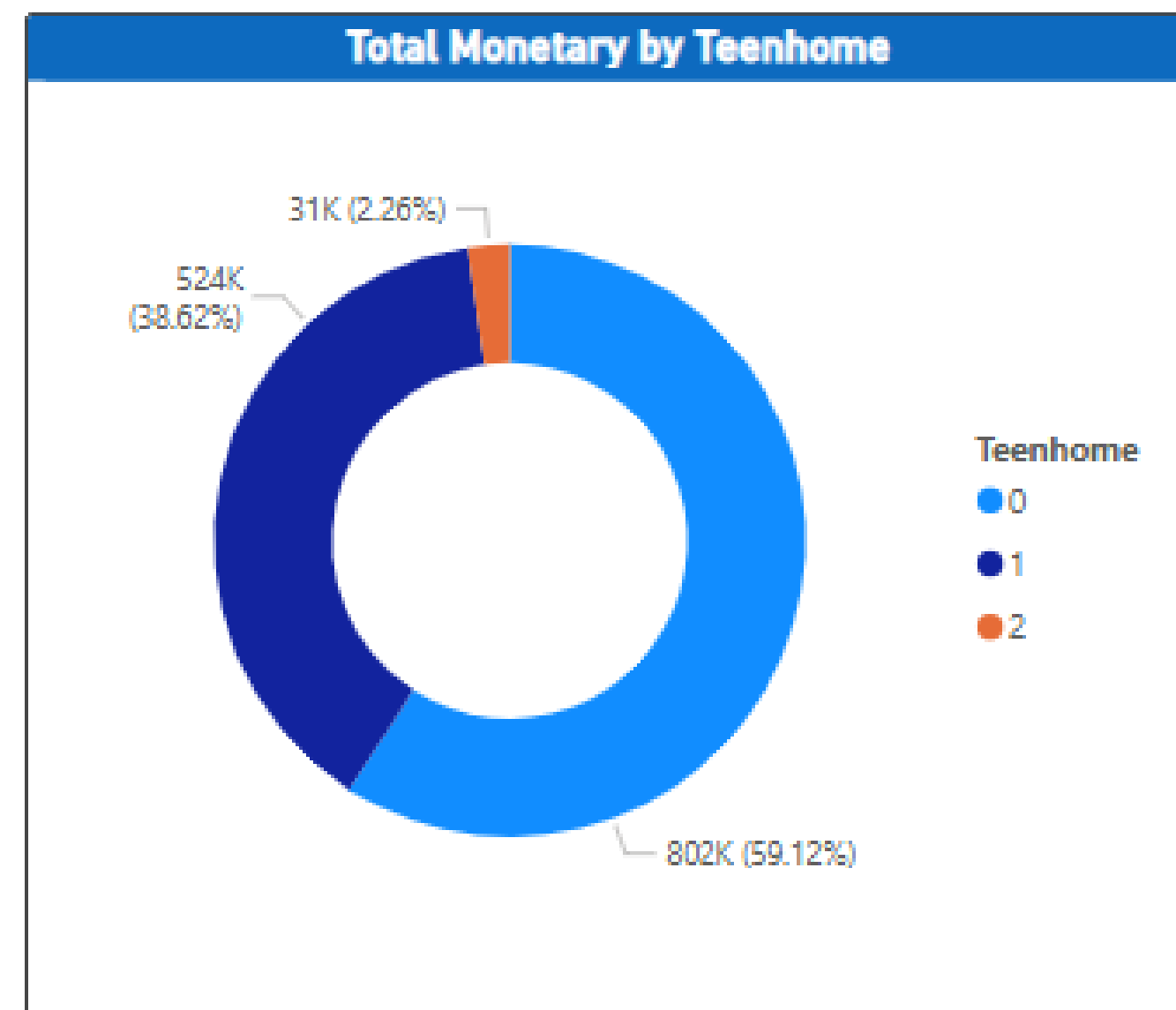
- There are significant fluctuations in total monetary value across the entire recency range.
- The highest peak appears around a recency value of 30, indicating a period when customers contributed the most to the total monetary value.



DASHBOARD (3)



Households without children (0) dominate the total monetary value with 85.88%, indicating that the majority of contributions come from individuals without children at home.



No Teenagers at Home (0) have the largest contribution, accounting for 59.12% (802K). This indicates that households without teenagers have the highest purchasing power.

Dashboard Marketing Campaign

Education

Select all

2n Cycle

Basic

Graduation

Master

PhD

1127

Count of ID

594M

Total Income

699K

Total Monetary

Total Monetary by Education

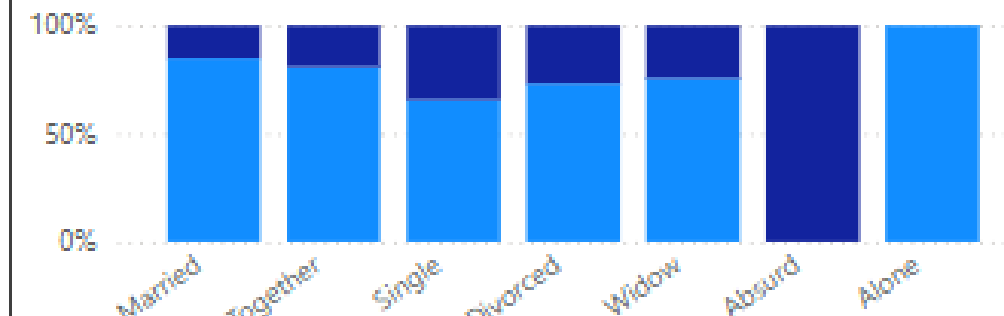
0.70M



Graduation
Education

Total Monetary by Marital_Status and Response

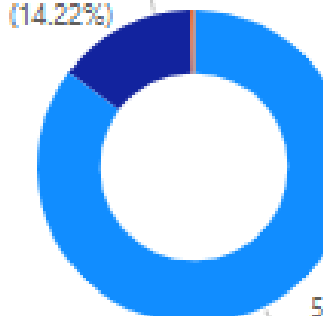
Response ● 0 ● 1



Marital_Status

Total Monetary by Kidhome

99K
(14.22%)



Kidhome

● 0

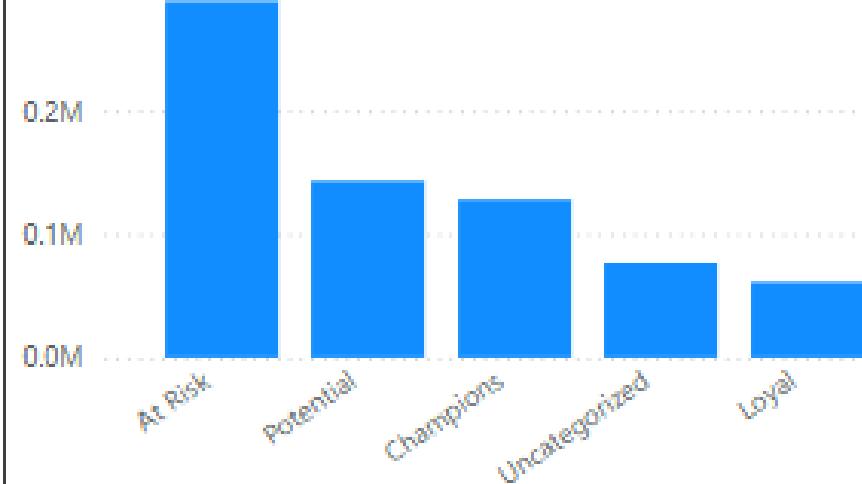
● 1

● 2

596K
(85.32%)

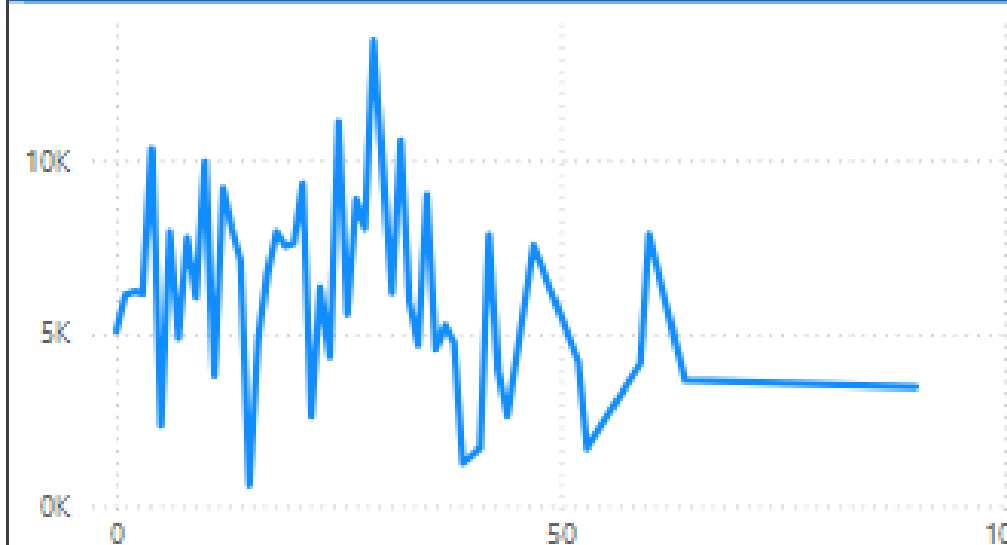
Total Monetary by Customer Segment

0.3M



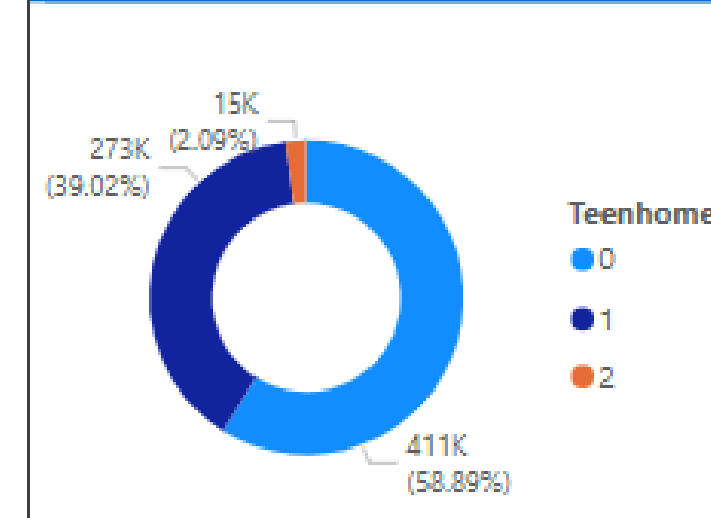
Customer Segment

Total Monetary by Recency



Recency

Total Monetary by Teenhome



Teenhome

● 0

● 1

● 2

273K
(39.02%)

15K
(2.09%)

411K
(58.89%)

The number of customers with a Graduation education level is 1,127, with a total income of 594 million and total spending of 699 thousand.

Total Monetary by Education

- The total monetary value for the Graduation group is 0.70M, indicating that customers with a Graduation education level have spent a total of 700,000.

Total Monetary by Marital Status and Response

- Segmentation by marital status shows that the "Married" and "Together" categories have the highest spending.
- However, their response rate to marketing campaigns is relatively low.

Total Monetary by Kidhome

- This chart illustrates total monetary value based on the number of children at home (Kidhome).
- Households with 0 children contribute the majority of the spending, around 596K (85.32%).



Total Monetary by Customer Segment

- The At Risk segment contributes the highest spending, approximately 0.3M.
- The Loyal and Uncategorized segments have lower spending contributions.

Total Monetary by Recency

- This chart shows significant fluctuations in spending, with several peaks.
- The highest spending peak occurs at Recency 29.

Total Monetary by Teenhome

- Households with 0 teenagers have higher spending, around 41K (58.89%).





A/B TESTING

Steps:

Hypothesis

H_0 : There is no significant correlation between Income and Monetary.
 H_a : There is a significant correlation between Income and Monetary.

P-value

- If p-value ≤ 0.05 , reject H_0
- If p-value > 0.05 , accept H_0

Statistical Test

- Pearson Correlation Test: Used if the data follows a normal distribution.
- Spearman Rank Correlation Test: Used if the data does not follow a normal distribution.





```
Shapiro-Wilk Test Stat: 0.7774129504529002  
P-Value: 5.606942962295885e-48  
The data is not normally distributed (reject H0).
```

INCOME

- $P\text{-value} > 0.05$
- Conclusion: Income is not normally distributed.

NORMALITY TEST

```
Shapiro-Wilk Test Stat: 0.8646703273311537  
P-Value: 2.7805851314631663e-40  
The data is not normally distributed (reject H0).
```

MONETARY

- $P\text{-value} > 0.05$
- Conclusion: Monetary is not normally distributed.

Thus, the statistical test used:
Spearman Rank Correlation Test



HYPOTHESIS TEST

```
Spearman's Rank Correlation Statistic: 0.8473181315399081  
P-Value: 0.0
```

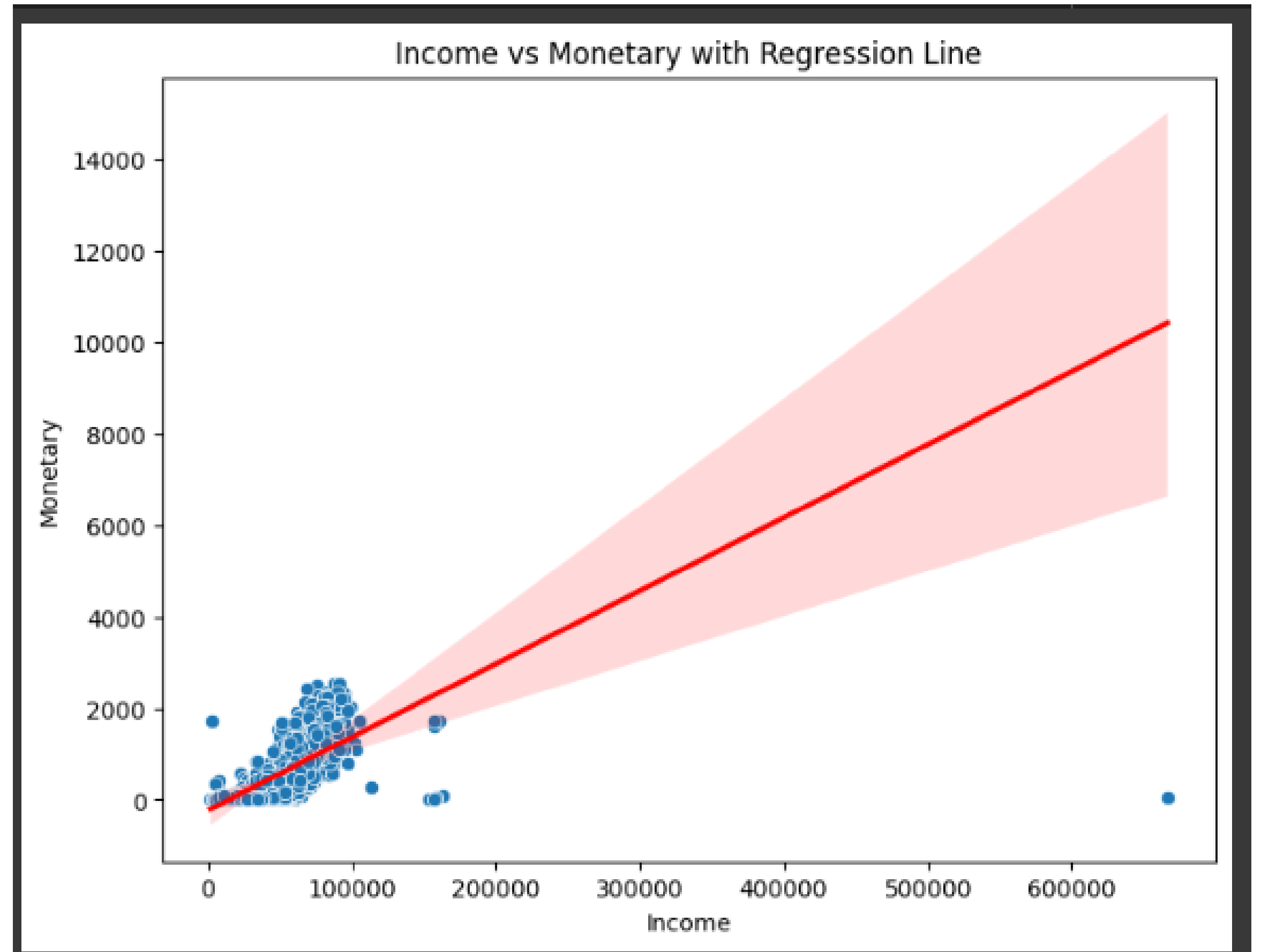
Test Result:

- $P\text{value} \leq 0,05$
- Conclusion: There is a significant correlation between Income and Monetary.

Interpretation:

Indicates a positive correlation.

There is a clear positive relationship between income and spending.





ACTION PLAN

Focus on:

- 1. Bachelor's Degree & At Risk Segment: Prioritize marketing efforts towards these high-spending groups.**
- 2. Loyalty Program for Loyal Segment: Develop exclusive rewards to retain long-term customers.**
- 3. Tailored Campaigns by Marital Status & Response: Adjust marketing strategies based on engagement patterns.**
- 4. Monitor Recency Fluctuations: Identify peak purchasing periods to optimize campaign timing.**
- 5. Target Households Without Kids & Teens: These groups have the highest purchasing power.**
- 6. Premium Products for High-Income Groups: Offer personalized, high-value products/services.**
- 7. Customer Response Segmentation: Actively target engaged individuals like "YOLO" with action-driven campaigns.**

About me

Hello. I am Arianty. I am a Biostatistician with a Master's degree from PH University of Indonesia and specialized expertise in data analysis, with proven experience in data entry, cleaning, merging, and analysis. I have contributed to impactful health initiatives at reputable organizations, including BKKBN, Yayasan KNCV Indonesia, and Dinas Kesehatan DKI Jakarta. Passionate about the data field, I bring over 4 years of experience as a Data Officer, specializing in managing and analyzing large datasets to derive actionable insights. I am skilled in tools such as Excel, SPSS, STATA, Tableau, Power BI, SQL, and Python and have a strong ability to turn complex data into meaningful outcomes.



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THANK YOU

